

MINA ABDEL-MASIH

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Education

North Carolina State University

Expected May 2028

Bachelor of Science in Computer Science (AI Concentration), Math Minor | GPA: 4.0/4.0

Raleigh, NC

- **Ranked #1 in Computer Science (306), Engineering (2,205), and Junior Class (7,793)**
- **Honors/Awards:** NCRMA Scholar, Henry B Rowe Jr. Scholar, Robert F. & Romaine S. Angel Scholar, Dean's List
- **Courses:** Data Structures & Algorithms, Machine Learning & Data Analysis, C & Software Tools, Software Development

Technical Skills

Programming Languages (Proficiency): Java, C, Python, LaTeX, HTML/CSS, JavaScript

Technologies: Git, GitHub, Microsoft Office, VSCode, Eclipse, Google Workspace, React.js, Claude Code, Postman, n8n, Jenkins

Certifications: ChatGPT Prompt Engineering for Developers, IBM Unleashing The Power of AI Agents, IBM Generative AI Essentials, Wells Fargo Software Engineering Job Simulation Certificate, IBM RAG for Enhanced AI Output

Work Experience

Zaki Technology

May 2026 – Present

Automation Developer Intern

Raleigh, NC

- Scoped an **AI workflow automation** for an enterprise client to replace a manual 2–3 hour daily data-entry process, analyzing the existing pipeline and architecting an email-ingestion, classification, and human-review system using **n8n**, the **Gemini API**, and **Google Workspace**
- Designed a **human-in-the-loop MVP** with a structured **JSON** classification schema (loss type, status, and confidence score), prioritizing reliability and review safeguards before any automated database writes
- Engineered an automated **email-parsing and labeling workflow** to ingest 20–30 daily alerts, extracting article titles, URLs, and snippets for downstream AI classification and structured **Google Sheets** output

North Carolina State University

Aug. 2024 – Dec. 2024

Information Security Research

Raleigh, NC

- Investigated privacy risks and mitigation strategies across 3 levels (individual, organizational, societal); synthesized insights from 6 peer-reviewed studies (2009–2021) on authentication (2FA/biometrics), encryption, and human-factor drivers (privacy fatigue, awareness, common mistakes), highlighting evidence that enabling 2FA reduces compromise risk by **≈99%**
- Engineered a **full-stack research showcase website** using **HTML, CSS, and JavaScript**, achieving 100% responsive design and smooth interactive transitions; leveraged AI-assisted development tools to accelerate design and debugging, demonstrating the ability to translate complex research into a fast, accessible product

Projects

Rooted | IBM x NCCU AI Hackathon | watsonx, LangChain, FastAPI, Python, PostgreSQL, Docker

March 2026

- * Built an **agentic AI therapy journaling app** at the IBM x NCCU Hackathon as **AI/ML lead**, owning the **IBM watsonx** layer — crisis routing, **sentiment scoring**, and follow-up generation via **LangChain** and **watsonx Orchestrate**
- * Engineered a **three-tier degradation strategy** (watsonx Orchestrate REST agent → per-tool **granite-13b-chat-v2** calls → deterministic Python pipeline) with an **AgentTrace** emitted in every API response, so the product degrades in output quality rather than failing when the model provider is unreachable
- * Built a **zero-LLM-latency crisis pathway** that classifies locally before persistence and auto-escalates flagged nodes, on a **role-based async REST API** (FastAPI, PostgreSQL, asyncpg, SQLAlchemy, JWT), containerized with **Docker Compose** health checks and auto-seeded demo data

Student Productivity Tool | Java 21, Spring Boot, JUnit 5, React, Vite, PostgreSQL, REST API, FullCalendar

May 2026

- * Built a **full-stack academic progress tracker** with a **Java 21 / Spring Boot REST API**, **React 19 + Vite** frontend, and **PostgreSQL** persistence, exposing **20 endpoints** for courses, assignments, notes, deadlines, and grade analytics
- * Engineered a **grade calculation engine** that computes weighted course grades, projects the scores needed on remaining assignments to reach target grades, flags unreachable targets, and classifies courses as on-track, at-risk, or critical
- * Designed a **React SPA** with **18 reusable components** across **6 routed pages** and a **FullCalendar deadline view** that distinguishes upcoming and past-due assignments backed by **Spring Data JPA** relationships

Access Control Compliance Engine | Java, JUnit, CSV I/O, Custom Data Structures

February 2026

- * Designed and built a **Java-based compliance tool** for an electronic medical records client, automatically parsing user activity logs against access control rule sets to classify events as violations, warnings, or compliant actions across datasets of 262,000+ records
- * Implemented **custom list-based Map and sorting algorithms from scratch** to enable multi-key sorted report generation (by username, timestamp, action, and resource), meeting strict sub-33 second performance benchmarks on large-scale inputs
- * Delivered the **full software development lifecycle** end-to-end: authored a formal proposal, produced a UML class diagram, wrote a structured system test plan, implemented the solution in Java, and validated correctness through **JUnit testing**

MinaAbdel-Masih.com | Portfolio Website | React, Vite, JavaScript, CSS

August 2026

- * Built a **React + Vite portfolio (13 components)** on a **data-driven content layer**, where projects, skills, and bio live in shared modules rendered by both the UI and the AI assistant, keeping a single source of truth
- * Engineered **two serverless backends** with zero infrastructure provisioning: a **Vercel function** fronting the **Gemini API** with the key held server-side and hardened by **per-IP rate limiting, token caps, and prompt-injection guards**, plus a **Google Apps Script** endpoint persisting contact submissions to **Sheets**
- * Implemented an interactive **3D skills visualization** with **Three.js + React Three Fiber**, code-split behind a lazy import so the 818 KB **WebGL** bundle never loads on mobile or under reduced-motion, with a lightweight fallback rendering instead

Student Organizations & Extracurriculars

Hackpack | Artificial Intelligence Club | NCSU App Dev Club | NCSU Oriental Orthodox Club (*Treasurer*) | UNC Coptic Student Organization | NCSU Arab Student Association | UNC Arab Student Association | NCSU Society of Sales Engineers